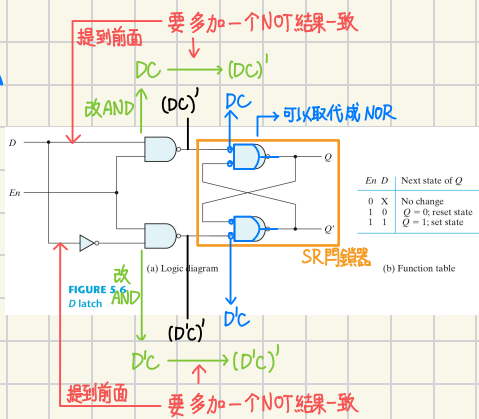
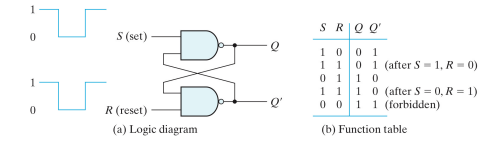




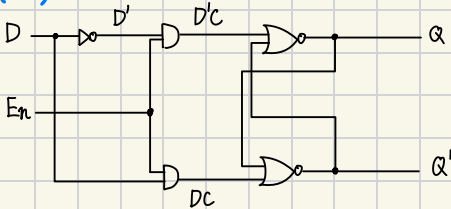
1.



SR 門鎖器: 由两个 NOR 或 NAND 交叉耦合。

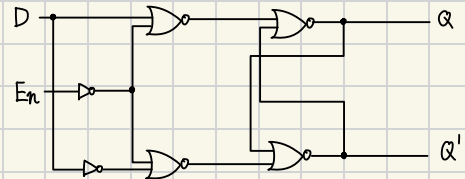


(a.)



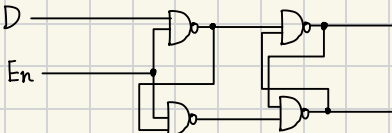
En	D	Q(t+1)
0	X	No Change
1	0	Q=0, reset state
1	1	Q=1, set state

(b)



En	D	Q(t+1)
0	X	No Change
1	0	Q=0, reset state
1	1	Q=1, set state

(c)



En	D	Q(t+1)
0	X	No Change
1	0	Q=0, reset state
1	1	Q=1, set state

2.

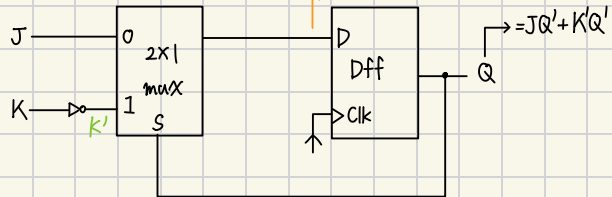
JK flip-flop 特徵表

J	K	Q(t+1)
0	0	Q(t) No Change
0	1	0 Reset State
1	0	1 Set State
1	1	Q'(t) Complement

$$D = JQ' + K'Q$$

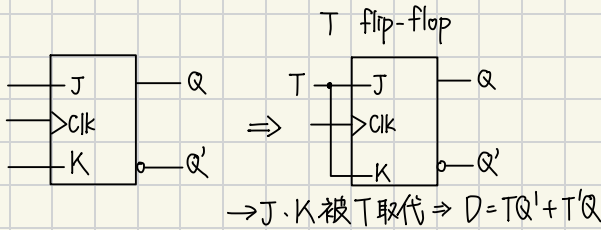
Q=0 時為 Q', mux 選擇 J

$JQ' + K'Q \Rightarrow Q=1$ 時是 Q, mux 選 K'



3.

JK 特徵方程式 $D = JQ' + K'Q$



T flip-flop 特徵表

T	Q(t)	Q(t+1)	
0	0	0	No Change
0	1	1	No Change
1	0	1	Complement
1	1	0	Complement

$$Q(t+1) = TQ'(t) + T'Q(t)$$

4.

(a)

A	B	Q(t+1)
0	0	Q(t) Complement
0	1	1
1	0	0
1	1	Q(t) No Change

(b)

Q(t)	Q(t+1)	A	B
0	0	1	x
0	1	0	x
1	0	x	0
1	1	x	1

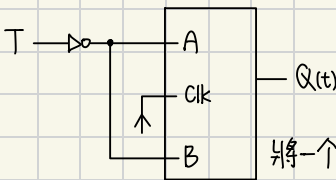
(c)

A	B	Q(t)	Q(t+1)
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

(d)

T flip-flop 特徵表

T	Q(t+1)	A B
0	No Change	1 1
1	Complement	0 0



A \ BQ(t)	00	01	11	10
0	1	0	1	1
1	0	0	1	0

$$Q(t+1) = A'Q'(t) + BQ(t)$$

5.

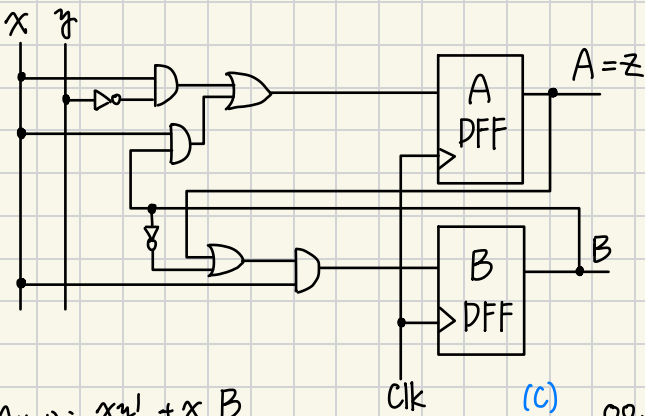
Truth Table 描述的是組合邏輯、State Table 則描述序項邏輯。

Characteristic Table 描述的是正反器的操作, Excitation Table 則描述特定狀態轉換下, 正反器各個輸入的取值。

Boolean Equation、State Equation、Characteristic Equation、Excitation Equation 這四種方程式則分別對應了上述四種表的代數表式形式。

6.

(a)



(b)

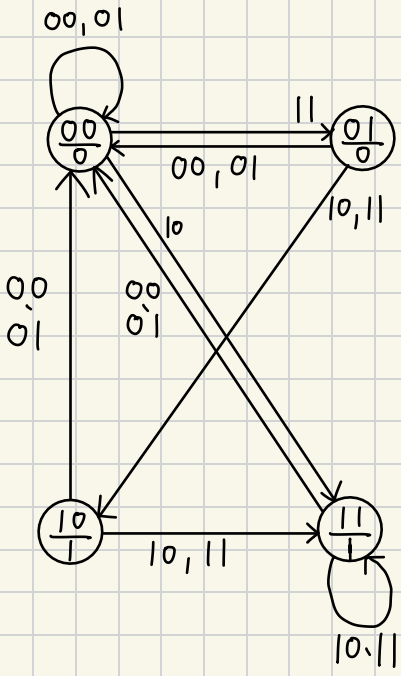
$$A(t+1) = xy' + xB$$

$$B(t+1) = xA + xB'$$

$$z = A$$

Present		Input		Next		Output
A	B	x	y	A	B	z
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	1	1	0
0	0	1	1	0	1	0
0	1	0	0	0	0	0
0	1	0	1	0	0	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	1	1	1
1	0	1	1	1	1	1
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1

(c)



7.

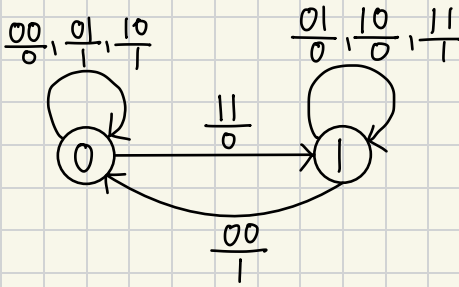
Present Q	Input x y	Next Q(t+1)	Output S
0	0 0	0	0
0	0 1	0	1
0	1 0	0	1
0	1 1	1	0
1	0 0	0	1
1	0 1	1	0
1	1 0	1	0
1	1 1	1	1

Q \ xy	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$S = x \oplus y \oplus Q$

Q \ xy	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$Q(t+1) = Qy + Qx + xy$



T	Q(t+1)
0	No Change
1	Complement

$T_A = A + B$ $T_B = A' + B$

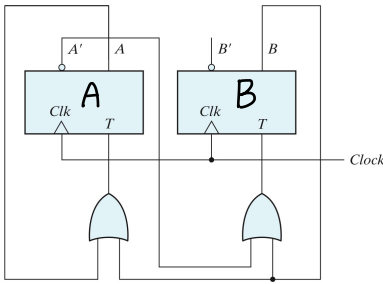
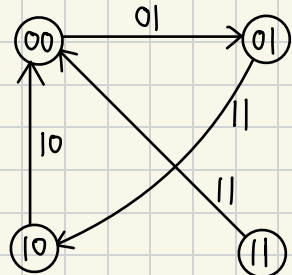


FIGURE P5.8

Present		Input		Next	
A	B	T _A	T _B	A	B
0	0	0	1	0	1
0	1	1	1	1	0
1	0	1	0	0	0
1	1	1	1	0	0

repeated sequence:
 $\boxed{00 \rightarrow 01 \rightarrow 10}$

這是一個 repeated sequence 為
 00, 01, 10 的計數器。



9.

(a)

$$J_A = x', K_A = B', J_B = A, K_B = x$$

$$A(t+1) = x'A' + BA$$

$$B(t+1) = AB' + x'B$$

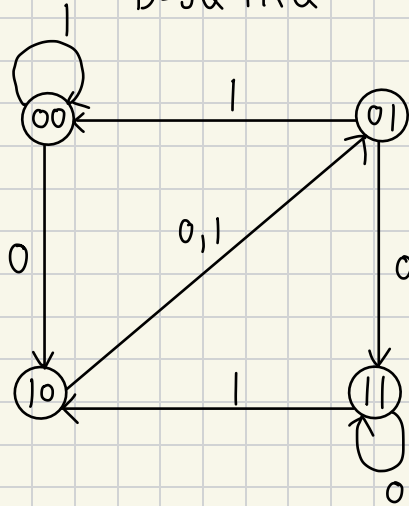
JK flip flop 特徵表

J	K	Q(t+1)	
0	0	Q(t)	No Change
0	1	0	Reset
1	0	1	Set
1	1	Q'(t)	Complement

$$D = JQ' + K'Q$$

(b)

Present		Input	Next	
A	B	x	A	B
0	0	0	1	0
0	0	1	0	0
0	1	0	1	1
0	1	1	0	0
1	0	0	0	1
1	0	1	0	1
1	1	0	1	1
1	1	1	1	0



10.

$$J_A = A'x + B'y, K_A = Bx'y', z = A'x' + B'y'$$

$$J_B = A'xy, K_B = A' + B'x$$

$$A(t+1) = (A'x + B'y)A' + (Bx'y')'A = A'x + A'B'y + (B' + x + y)A$$

$$= \underline{A'x} + A'B'y + AB' + \underline{Ax} + Ay = x(A+A') + B'(A+A'y) + Ay$$

$$= x + B'(A+y) + Ay = x + AB' + B'y + Ay$$

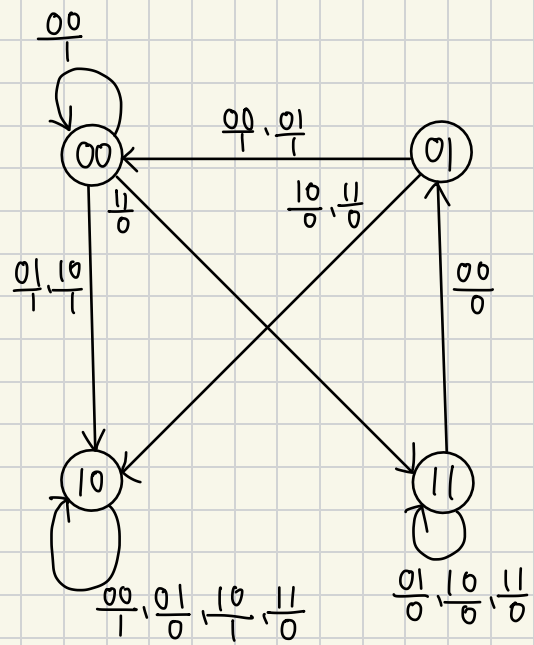
$$B(t+1) = (A'xy)B + (A' + B'x)'B = A'B'xy + A \underbrace{(B+x)B}_{\text{吸收律=B}} = A'B'xy + AB$$

10.

$$A(t+1) = x + AB' + B'y + Ay \quad z = A'x' + B'y'$$

$$B(t+1) = A'B'xy + AB$$

Present		Input		Next		Output
A	B	x	y	A	B	z
0	0	0	0	0	0	1
0	0	0	1	1	0	1
0	0	1	0	1	0	1
0	0	1	1	1	1	0
0	1	0	0	0	0	1
0	1	0	1	0	0	1
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	1	0	1
1	0	0	1	1	0	0
1	0	1	0	1	0	1
1	0	1	1	1	0	0
1	1	0	0	0	1	0
1	1	0	1	1	1	0
1	1	1	0	1	1	0
1	1	1	1	1	1	0



11,
(a)

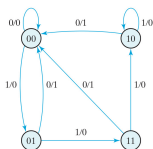


FIGURE 5.16
State diagram of the circuit of Fig. 5.15

Present 00 01 11 00 00 01 00 01 00 00 01 11 10 00 01
Input 1 1 0 0 1 0 1 0 0 1 1 1 0 1 0
Output 0 0 1 0 0 1 0 1 0 0 0 0 1 0 1
Next 01 11 00 00 01 00 01 00 00 01 11 10 00 01 00

Output Sequence: 0010 0101 0000 101

(b) a=00, b=01, c=10, d=11

Present	Next		Output	
	x=0	x=1	x=0	x=1
a	a	b	0	0
b	a	d	1	0
c	a	c	1	0
d	a	c	1	0

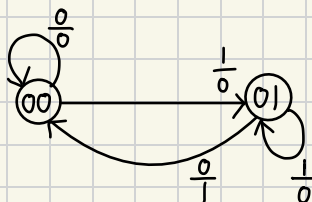
Present	Next		Output	
	x=0	x=1	x=0	x=1
a	a	b	0	0
b	a	c	1	0
c	a	c	1	0

Present	Next		Output	
	x=0	x=1	x=0	x=1
a	a	b	0	0
b	a	b	1	0

Present	Next		Output	
	x=0	x=1	x=0	x=1
00	00	01	0	0
01	00	01	1	0

Present 00 01 01 00 00 01 00 01 00 00 01 01 01 00 01
Input 1 1 0 0 1 0 1 0 0 1 1 1 0 1 0
Output 0 0 1 0 0 1 0 1 0 0 0 0 1 0 1
Next 01 01 00 00 01 00 01 00 00 01 01 01 00 01 00

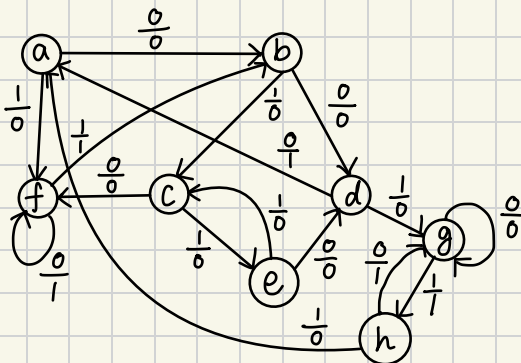
Output Sequence 0010 0101 0000 101



12,

Present State	Next State		Output	
	x=0	x=1	x=0	x=1
a	f	b	0	0
b	d	c	0	0
c	f	b	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	d	0	1
h	g	a	1	0

Present	Next		Output	
	x=0	x=1	x=0	x=1
a	f	b	0	0
b	d	c	0	0
c	f	b	0	0
d	g	a	1	0
f	f	b	1	1
g	g	d	0	1

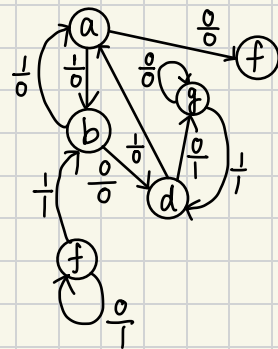


12.

Present	Next		Output	
	$x=0$	$x=1$	$x=0$	$x=1$
a	f	b	0	0
b	d	c	0	0
c	f	b	0	0
d	g	a	1	0
f	f	b	1	1
g	g	d	0	1



Present	Next		Output	
	$x=0$	$x=1$	$x=0$	$x=1$
a	f	b	0	0
b	d	a	0	0
d	g	a	1	0
f	f	b	1	1
g	g	d	0	1



13.

Present State	Next State		Output	
	$x=0$	$x=1$	$x=0$	$x=1$
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

Present	a	f	b	c	e	d	g	h	g	g	h
Input	0	1	1	1	0	0	1	0	0	1	1
Output	0	1	0	0	0	1	1	1	0	1	0
Next	f	b	c	e	d	g	h	g	g	h	a

Output Sequence: 0100 0111 010

Present	Next		Output	
	$x=0$	$x=1$	$x=0$	$x=1$
a	f	b	0	0
b	d	a	0	0
d	g	a	1	0
f	f	b	1	1
g	g	d	0	1

Present	a	f	b	a	b	d	g	d	g	g	d
Input	0	1	1	1	0	0	1	0	0	1	1
Output	0	1	0	0	0	1	1	1	0	1	0
Next	f	b	a	b	d	g	d	g	g	d	a

Output Sequence: 0100 0111 010

14.

Table 5.8
Reduced State Table

Present State	Next State		Output	
	$x=0$	$x=1$	$x=0$	$x=1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	d	0	1
e	a	d	0	1

Table 5.9
Three Possible Binary State Assignments

State	Assignment 1, Binary	Assignment 2, Gray Code	Assignment 3, One-Hot
a	000	000	0001
b	001	001	0010
c	010	011	0010
d	011	010	0100
e	100	110	1000

Present	Next		Output	
	$x=0$	$x=1$	$x=0$	$x=1$
000	000	001	0	0
001	011	010	0	0
011	000	010	0	0
010	110	010	0	1
110	000	010	0	1

15.

Present	Ext J'	Input K	Int J	Input K	Next	JK flip flop 特徵表
0	0	0	1	0	1 Set	J K Q(t+1)
0	0	1	1	1	1 Comp'	0 0 Q(t) No Change
0	1	0	0	0	0 x	0 1 0 Reset
0	1	1	0	1	0 Reset	1 0 1 Set
1	0	0	1	0	1 Set	1 1 Q'(t) Complement
1	0	1	1	1	0 Comp'	
1	1	0	0	0	1 x	
1	1	1	0	1	0 Reset	

Q \ J'K	00	01	11	10
0	1	1	0	0
1	1	0	0	1

$$Q(t+1) = JQ' + K'Q$$

16. (a)

Present	Input	Next
A B	x	A B
0 0	0	0 0
0 0	1	0 1
0 1	0	0 1
0 1	1	1 1
1 0	0	1 0
1 0	1	0 0
1 1	0	1 1
1 1	1	1 0

$$D_A = Ax' + Bx$$

A \ Bx	00	01	11	10
0	0	0	1	0
1	1	0	1	1

$$D_B = A'x + Bx'$$

A \ Bx	00	01	11	10
0	0	1	1	1
1	0	0	0	1

(b)

Present	Input	Next
A B	x	A B
0 0	0	0 0
0 0	1	1 1
0 1	0	0 1
0 1	1	1 0
1 0	0	1 0
1 0	1	0 0
1 1	0	1 1
1 1	1	0 1

$$D_A = A'x + Ax'$$

A \ Bx	00	01	11	10
0	0	1	1	0
1	1	0	0	1

$$D_B = A'B'x + AB + Bx'$$

A \ Bx	00	01	11	10
0	0	1	0	1
1	0	0	1	1

17.

Present	Input	Next	Output
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	0

$$D_A = A + X$$

$$y = A \oplus X$$

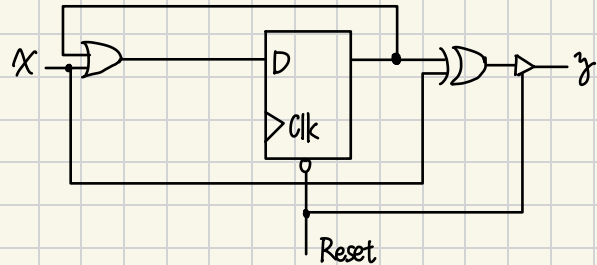
A \ X	0	1
0	0	1
1	1	1

A \ X	0	1
0	0	1
1	1	0

D flip flop 特徵表

D	$Q(t+1)$	
0	0	Reset
1	1	Set

看到第1个1: 1 → 之後所有輸入取NOT
沒看到第1个1: 0 → 直接輸出



18

Present		Input		Next			
A	B	E	F	A	B	J_A	K_A
0	0	0	0	0	0	0	X
0	0	0	1	0	0	0	X
0	0	1	0	1	1	1	X
0	0	1	1	0	1	0	X
0	1	0	0	0	1	0	X
0	1	0	1	0	1	0	X
0	1	1	0	0	1	0	X
0	1	1	1	1	0	1	X
1	0	0	0	1	0	X	0
1	0	0	1	1	0	X	0
1	0	1	0	0	1	X	1
1	0	1	1	1	1	X	0
1	1	0	0	1	1	X	0
1	1	0	1	1	1	X	0
1	1	1	0	1	0	X	0
1	1	1	1	0	0	X	1

JK 特徵表

J	K	$Q(t+1)$
0	0	Q
0	1	0 Reset
1	0	1 Set
1	1	Q'

JK 激勵表

Q	$Q(t+1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

$$J_A = BEF + B'EF'$$

$$K_A = BEF + B'EF'$$

AB \ EF	00	01	11	10
00	0	0	0	1
01	0	0	1	0
11	X	X	X	X
10	X	X	X	X

AB \ EF	00	01	11	10
00	X	X	X	X
01	X	X	X	X
11	0	0	1	0
10	0	0	0	1

$$J_B = E$$

$$K_B = E$$

AB \ EF	00	01	11	10
00	0	0	1	1
01	X	X	X	X
11	X	X	X	X
10	0	0	1	1

AB \ EF	00	01	11	10
00	X	X	X	X
01	0	0	1	1
11	0	0	1	1
10	X	X	X	X

19.

(a)

Present			Input	Next	Output
A	B	C	x	ABC	y
0	0	0	0	011	0
0	0	0	1	100	1
0	0	1	0	001	0
0	0	1	1	100	1
0	1	0	0	010	0
0	1	0	1	000	1
0	1	1	0	001	0
0	1	1	1	010	1
1	0	0	0	010	0
1	0	0	1	011	1
1	0	1	0	X	
1	0	1	1		
1	1	0	0		
1	1	0	1		
1	1	1	0		
1	1	1	1	X	
1	1	1	0		
1	1	1	1		
1	1	1	0		
1	1	1	1		

$$D_A = A'B'x$$

AB \ Cx	00	01	11	10
00	0	1	1	0
01	0	0	0	0
11	x	x	x	x
10	0	0	x	x

$$D_B = A + Cx' + Bcx$$

AB \ Cx	00	01	11	10
00	1	0	0	0
01	1	0	1	0
11	x	x	x	x
10	1	1	x	x

$$D_C = Ax + Cx' + AB'x' \quad y = x$$

AB \ Cx	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	x	x	x	x
10	0	1	x	x

AB \ Cx	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	x	x	x	x
10	0	1	x	x

(b)

J _A	K _A	J _B	K _B	J _C	K _C
0	x	1	x	1	x
1	x	0	x	0	x
0	x	0	x	x	0
1	x	0	x	x	1
0	x	x	0	0	x
0	x	x	1	0	x
0	x	x	1	x	0
0	x	x	0	x	1
x	1	1	x	0	x
x	1	1	x	1	x

JK 特徵表

J	K	Q(t+1)
0	0	Q
0	1	0 Reset
1	0	1 Set
1	1	Q'

JK 激勵表

Q	Q(t+1)	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

JK

J	K	$Q(t+1)$	Q	$Q(t+1)$	J	K
0	0	No Change	0	0	0	X
0	1	0 Reset	0	1	1	X
1	0	1 Set	1	0	X	1
1	1	Comp'	1	1	X	0

$$D = JQ' + K'Q$$

D:

D	$Q(t+1)$
0	0 Reset
1	1 Set

T:

T	$Q(t+1)$	$Q(t)$	$Q(t+1)$	T
0	$Q(t)$ No Change	0	0	0
1	$Q'(t)$ Comp'	0	1	1
		1	0	1
		1	1	0

$$D = TQ' + T'Q$$

